



February 04, 2026

Cindy Tomaszewski
City of Cadillac
1121 Plett Rd
Cadillac, MI 49601

RE: Project: Effluent PFAS
Pace Project No.: 50422857

Dear Cindy Tomaszewski:

Enclosed are the analytical results for sample(s) received by the laboratory on January 22, 2026. The results relate only to the samples included in this report. Results reported herein conform to the applicable TNI/NELAC Standards and the laboratory's Quality Manual, where applicable, unless otherwise noted in the body of the report.

The test results provided in this final report were generated by each of the following laboratories within the Pace Network:

- Pace Analytical Services - Minneapolis

If you have any questions concerning this report, please feel free to contact me.

Sincerely,

A handwritten signature in blue ink that reads "Brian Hall".

Brian Hall
brian.hall@pacelabs.com
(616)975-4500
Project Manager

Enclosures

cc:



REPORT OF LABORATORY ANALYSIS

This report shall not be reproduced, except in full,
without the written consent of Pace Analytical Services, LLC.

Table of Contents

Sections:

Section 1: Cert Page	3
Section 2: Sample Summary	4
Section 3: Sample Analyte	5
Section 4: Summary of Hits	6
Section 5: Results	7
Section 6: Quality Control	9
Section 7: Qualifiers	13
Section 8: QC Cross Reference	14
Section 9: Chain of Custody	15
9.1: Chain of Custody	15

1

2

3

4

5

6

7

8

9



CERTIFICATIONS

Project: Effluent PFAS

Pace Project No.: 50422857

Pace Analytical Services, LLC - Minneapolis MN

1700 Elm Street SE, Minneapolis, MN 55414
Alaska Contaminated Sites Certification #: 17-009
Arizona Certification #: AZ0014
Arkansas WW Certification #: 88-0680
Colorado Certification #: MN00064
DoD Certification via A2LA #: 2926.01
Florida Certification #: E87605
Idaho Certification #: MN00064
Indiana Certification #: C-MN-01
ISO/IEC 17025 Certification via A2LA #: 2926.01
Kentucky DW Certification #: 90062
Louisiana DEQ Certification #: AI-03086
Maine Certification #: MN00064
Michigan Certification #: 9909
Minnesota Dept of Ag Approval: via MN 027-053-137
Mississippi Certification #: MN00064
Montana Certification #: CERT0092
Nevada Certification #: MN00064
New Jersey Certification #: MN002
North Carolina DW Certification #: 27700
North Dakota Certification via A2LA #: R-036
Ohio DW Certification #: 41244
Oklahoma Certification #: 9507
Oregon Secondary Certification #: MN200001
Puerto Rico Certification #: MN00064
Tennessee Certification #: TN02818
Utah Certification #: MN00064
Virginia Certification #: 460163
West Virginia DEP Certification #: 382
Wisconsin Certification #: 999407970
USDA Permit #: P330-19-00208

Alabama Certification #: 40770
Alaska DW Certification #: MN00064
Arkansas DW Certification #: MN00064
California Certification #: 2929
Connecticut Certification #: PH-0256
EPA Region 8 Tribal Water Systems+Wyoming DW Certification #: via MN 027-053-137
Georgia Certification #: 959
Illinois Certification #: 200011
Iowa Certification #: 368
Kansas Certification #: E-10167
Kentucky WW Certification #: 90062
Louisiana DW Certification #: MN00064
Maryland Certification #: 322
Minnesota Certification #: 027-053-137
Minnesota Petrofund Registration #: 1240
Missouri Certification #: 10100
Nebraska Certification #: NE-OS-18-06
New Hampshire Certification #: 2081
New York Certification #: 11647
North Carolina WW Certification #: 530
North Dakota Certification via MN #: R-036
Ohio VAP Certification (1700) #: CL101
Oregon Primary Certification #: MN300001
Pennsylvania Certification #: 68-00563
South Carolina Certification #: 74003001
Texas Certification #: T104704192
Vermont Certification #: VT-027053137
Washington Certification #: C486
West Virginia DW Certification #: 9952 C
Wyoming UST Certification via A2LA #: 2926.01

REPORT OF LABORATORY ANALYSIS

This report shall not be reproduced, except in full,
without the written consent of Pace Analytical Services, LLC.

Date: 02/04/2026 03:20 PM



SAMPLE SUMMARY

Project: Effluent PFAS
Pace Project No.: 50422857

Lab ID	Sample ID	Matrix	Date Collected	Date Received
50422857001	Effluent	Water	01/21/26 13:05	01/22/26 11:15

REPORT OF LABORATORY ANALYSIS

This report shall not be reproduced, except in full,
without the written consent of Pace Analytical Services, LLC.



SAMPLE ANALYTE COUNT

Project: Effluent PFAS
Pace Project No.: 50422857

Lab ID	Sample ID	Method	Analysts	Analytes Reported	Laboratory
50422857001	Effluent	EPA 1633	MJL	65	PASI-M

PASI-M = Pace Analytical Services - Minneapolis

REPORT OF LABORATORY ANALYSIS

This report shall not be reproduced, except in full,
without the written consent of Pace Analytical Services, LLC.



SUMMARY OF DETECTION

Project: Effluent PFAS

Pace Project No.: 50422857

Lab Sample ID	Client Sample ID					
Method	Parameters	Result	Units	Report Limit	Analyzed	Qualifiers
50422857001	Effluent					
EPA 1633	NMeFOSAA	0.50J	ng/L	1.6	01/28/26 22:58	
EPA 1633	PFBS	5.1	ng/L	1.6	01/28/26 22:58	
EPA 1633	PFHxA	4.8	ng/L	1.6	01/28/26 22:58	
EPA 1633	PFBA	3.3J	ng/L	6.3	01/28/26 22:58	
EPA 1633	PFPeA	1.0J	ng/L	3.1	01/28/26 22:58	
EPA 1633	PFHpA	0.39J	ng/L	1.6	01/28/26 22:58	
EPA 1633	PFHxS	0.65J	ng/L	1.6	01/28/26 22:58	I
EPA 1633	PFNA	0.33J	ng/L	1.6	01/28/26 22:58	
EPA 1633	PFOS	1.1J	ng/L	1.6	01/28/26 22:58	
EPA 1633	PFOA	1.2J	ng/L	1.6	01/28/26 22:58	

REPORT OF LABORATORY ANALYSIS

This report shall not be reproduced, except in full,
without the written consent of Pace Analytical Services, LLC.

Date: 02/04/2026 03:20 PM



ANALYTICAL RESULTS

Project: Effluent PFAS

Pace Project No.: 50422857

Sample: Effluent		Lab ID: 50422857001		Collected: 01/21/26 13:05		Received: 01/22/26 11:15		Matrix: Water	
Parameters	Results	Units	PQL	MDL	DF	Prepared	Analyzed	CAS No.	Qual
EPA 1633 Water									
Analytical Method: EPA 1633 Preparation Method: EPA 1633									
Pace Analytical Services - Minneapolis									
11CI-PF3OUdS	<1.3	ng/L	6.3	1.3	1	01/28/26 07:57	01/28/26 22:58	763051-92-9	
3:3 FTCA	<2.0	ng/L	7.9	2.0	1	01/28/26 07:57	01/28/26 22:58	356-02-5	
4:2 FTS	<1.5	ng/L	6.3	1.5	1	01/28/26 07:57	01/28/26 22:58	757124-72-4	
5:3 FTCA	<5.3	ng/L	39.3	5.3	1	01/28/26 07:57	01/28/26 22:58	914637-49-3	
6:2 FTS	<1.5	ng/L	6.3	1.5	1	01/28/26 07:57	01/28/26 22:58	27619-97-2	
7:3 FTCA	<5.1	ng/L	39.3	5.1	1	01/28/26 07:57	01/28/26 22:58	812-70-4	
8:2 FTS	<2.4	ng/L	6.3	2.4	1	01/28/26 07:57	01/28/26 22:58	39108-34-4	
9CI-PF3ONS	<1.1	ng/L	6.3	1.1	1	01/28/26 07:57	01/28/26 22:58	756426-58-1	
ADONA	<0.76	ng/L	6.3	0.76	1	01/28/26 07:57	01/28/26 22:58	919005-14-4	
HFPO-DA	<0.94	ng/L	6.3	0.94	1	01/28/26 07:57	01/28/26 22:58	13252-13-6	
NEtFOSAA	<0.42	ng/L	1.6	0.42	1	01/28/26 07:57	01/28/26 22:58	2991-50-6	
NEtFOSA	<0.31	ng/L	1.6	0.31	1	01/28/26 07:57	01/28/26 22:58	4151-50-2	
NEtFOSE	<3.7	ng/L	15.7	3.7	1	01/28/26 07:57	01/28/26 22:58	1691-99-2	
NFDHA	<0.80	ng/L	3.1	0.80	1	01/28/26 07:57	01/28/26 22:58	151772-58-6	
NMeFOSAA	0.50J	ng/L	1.6	0.47	1	01/28/26 07:57	01/28/26 22:58	2355-31-9	
NMeFOSA	<0.52	ng/L	1.6	0.52	1	01/28/26 07:57	01/28/26 22:58	31506-32-8	
NMeFOSE	<2.8	ng/L	15.7	2.8	1	01/28/26 07:57	01/28/26 22:58	24448-09-7	
PFBS	5.1	ng/L	1.6	0.30	1	01/28/26 07:57	01/28/26 22:58	375-73-5	
PFDA	<0.35	ng/L	1.6	0.35	1	01/28/26 07:57	01/28/26 22:58	335-76-2	
PFHxA	4.8	ng/L	1.6	0.30	1	01/28/26 07:57	01/28/26 22:58	307-24-4	
PFBA	3.3J	ng/L	6.3	1.2	1	01/28/26 07:57	01/28/26 22:58	375-22-4	
PFDS	<0.46	ng/L	1.6	0.46	1	01/28/26 07:57	01/28/26 22:58	335-77-3	
PFDoS	<0.33	ng/L	1.6	0.33	1	01/28/26 07:57	01/28/26 22:58	79780-39-5	
PFEESA	<0.79	ng/L	3.1	0.79	1	01/28/26 07:57	01/28/26 22:58	113507-82-7	
PFHpS	<0.41	ng/L	1.6	0.41	1	01/28/26 07:57	01/28/26 22:58	375-92-8	
PFMBA	<0.46	ng/L	3.1	0.46	1	01/28/26 07:57	01/28/26 22:58	863090-89-5	
PFMPA	<0.59	ng/L	3.1	0.59	1	01/28/26 07:57	01/28/26 22:58	377-73-1	
PFNS	<0.44	ng/L	1.6	0.44	1	01/28/26 07:57	01/28/26 22:58	68259-12-1	
PFOSA	<0.30	ng/L	1.6	0.30	1	01/28/26 07:57	01/28/26 22:58	754-91-6	
PFPeA	1.0J	ng/L	3.1	0.44	1	01/28/26 07:57	01/28/26 22:58	2706-90-3	
PFPeS	<0.25	ng/L	1.6	0.25	1	01/28/26 07:57	01/28/26 22:58	2706-91-4	
PFDoA	<0.33	ng/L	1.6	0.33	1	01/28/26 07:57	01/28/26 22:58	307-55-1	
PFHpA	0.39J	ng/L	1.6	0.35	1	01/28/26 07:57	01/28/26 22:58	375-85-9	
PFHxS	0.65J	ng/L	1.6	0.42	1	01/28/26 07:57	01/28/26 22:58	355-46-4	I
PFNA	0.33J	ng/L	1.6	0.31	1	01/28/26 07:57	01/28/26 22:58	375-95-1	
PFOS	1.1J	ng/L	1.6	0.52	1	01/28/26 07:57	01/28/26 22:58	1763-23-1	
PFOA	1.2J	ng/L	1.6	0.45	1	01/28/26 07:57	01/28/26 22:58	335-67-1	
PFTeDA	<0.26	ng/L	1.6	0.26	1	01/28/26 07:57	01/28/26 22:58	376-06-7	
PFTrDA	<0.31	ng/L	1.6	0.31	1	01/28/26 07:57	01/28/26 22:58	72629-94-8	
PFUnA	<0.34	ng/L	1.6	0.34	1	01/28/26 07:57	01/28/26 22:58	2058-94-8	
Surrogates									
13C2-PFDoA (S)	71	%	10-130		1	01/28/26 07:57	01/28/26 22:58		
13C3HFPO-DA (S)	67	%	40-130		1	01/28/26 07:57	01/28/26 22:58		
13C3-PFBS (S)	71	%	40-135		1	01/28/26 07:57	01/28/26 22:58		

REPORT OF LABORATORY ANALYSIS

This report shall not be reproduced, except in full,
without the written consent of Pace Analytical Services, LLC.

Date: 02/04/2026 03:20 PM



ANALYTICAL RESULTS

Project: Effluent PFAS

Pace Project No.: 50422857

Sample: Effluent		Lab ID: 50422857001		Collected: 01/21/26 13:05		Received: 01/22/26 11:15		Matrix: Water	
Parameters	Results	Units	PQL	MDL	DF	Prepared	Analyzed	CAS No.	Qual
Surrogates									
13C3-PFHxS (S)	68	%.	40-130		1	01/28/26 07:57	01/28/26 22:58		
13C4-PFBA (S)	76	%.	5-130		1	01/28/26 07:57	01/28/26 22:58		
13C4-PFHpA (S)	71	%.	40-130		1	01/28/26 07:57	01/28/26 22:58		
13C5-PFHxA (S)	73	%.	40-130		1	01/28/26 07:57	01/28/26 22:58		
13C5-PFPeA (S)	71	%.	40-130		1	01/28/26 07:57	01/28/26 22:58		
13C6-PFDA (S)	73	%.	40-130		1	01/28/26 07:57	01/28/26 22:58		
13C8-PFOA (S)	71	%.	40-130		1	01/28/26 07:57	01/28/26 22:58		
13C8-PFOS (S)	73	%.	40-130		1	01/28/26 07:57	01/28/26 22:58		
13C8-PFOSA (S)	69	%.	40-130		1	01/28/26 07:57	01/28/26 22:58		
13C9-PFNA (S)	73	%.	40-130		1	01/28/26 07:57	01/28/26 22:58		
d3-MeFOSAA (S)	78	%.	40-170		1	01/28/26 07:57	01/28/26 22:58		
d3-NMeFOSA (S)	49	%.	10-130		1	01/28/26 07:57	01/28/26 22:58		
d5-EtFOSAA (S)	79	%.	25-135		1	01/28/26 07:57	01/28/26 22:58		
d5-NEtFOSA (S)	50	%.	10-130		1	01/28/26 07:57	01/28/26 22:58		
d7-NMeFOSE (S)	82	%.	10-130		1	01/28/26 07:57	01/28/26 22:58		
d9-NEtFOSE (S)	79	%.	10-130		1	01/28/26 07:57	01/28/26 22:58		
13C2-PFTA (S)	60	%.	10-130		1	01/28/26 07:57	01/28/26 22:58		
13C7-PFUdA (S)	73	%.	30-130		1	01/28/26 07:57	01/28/26 22:58		
13C24:2FTS (S)	106	%.	40-200		1	01/28/26 07:57	01/28/26 22:58		
13C26:2FTS (S)	121	%.	40-200		1	01/28/26 07:57	01/28/26 22:58		
13C28:2FTS (S)	128	%.	40-300		1	01/28/26 07:57	01/28/26 22:58		
13C3-PFPPrA (S)	53	%.	5-130		1	01/28/26 07:57	01/28/26 22:58		

REPORT OF LABORATORY ANALYSIS

This report shall not be reproduced, except in full,
without the written consent of Pace Analytical Services, LLC.

Date: 02/04/2026 03:20 PM



QUALITY CONTROL DATA

Project: Effluent PFAS

Pace Project No.: 50422857

QC Batch 1049892

QC Batch Method: EPA 1633

Analysis Method: EPA 1633

Analysis Description: EPA 1633 Water

Laboratory: Pace Analytical Services - Minneapolis

Associated Lab Samples: 50422857001

METHOD BLANK: 5460850

Matrix: Water

Associated Lab Samples: 50422857001

Parameter	Units	Blank Result	Reporting Limit	MDL	Analyzed	Qualifiers
11CI-PF3OUdS	ng/L	<1.3	6.4	1.3	01/28/26 20:47	
3:3 FTCA	ng/L	<2.1	8.0	2.1	01/28/26 20:47	
4:2 FTS	ng/L	<1.5	6.4	1.5	01/28/26 20:47	
5:3 FTCA	ng/L	<5.4	40.0	5.4	01/28/26 20:47	
6:2 FTS	ng/L	<1.5	6.4	1.5	01/28/26 20:47	
7:3 FTCA	ng/L	<5.2	40.0	5.2	01/28/26 20:47	
8:2 FTS	ng/L	<2.4	6.4	2.4	01/28/26 20:47	
9CI-PF3ONS	ng/L	<1.2	6.4	1.2	01/28/26 20:47	
ADONA	ng/L	<0.78	6.4	0.78	01/28/26 20:47	
HFPO-DA	ng/L	<0.96	6.4	0.96	01/28/26 20:47	
NEtFOSA	ng/L	<0.32	1.6	0.32	01/28/26 20:47	
NEtFOSAA	ng/L	<0.43	1.6	0.43	01/28/26 20:47	
NEtFOSE	ng/L	<3.8	16.0	3.8	01/28/26 20:47	
NFDHA	ng/L	<0.81	3.2	0.81	01/28/26 20:47	
NMeFOSA	ng/L	<0.53	1.6	0.53	01/28/26 20:47	
NMeFOSAA	ng/L	<0.48	1.6	0.48	01/28/26 20:47	
NMeFOSE	ng/L	<2.9	16.0	2.9	01/28/26 20:47	
PFBA	ng/L	<1.3	6.4	1.3	01/28/26 20:47	
PFBS	ng/L	<0.30	1.6	0.30	01/28/26 20:47	
PFDA	ng/L	<0.36	1.6	0.36	01/28/26 20:47	
PFDoA	ng/L	<0.34	1.6	0.34	01/28/26 20:47	
PFDoS	ng/L	<0.34	1.6	0.34	01/28/26 20:47	
PFDS	ng/L	<0.47	1.6	0.47	01/28/26 20:47	
PFEESA	ng/L	<0.80	3.2	0.80	01/28/26 20:47	
PFHpA	ng/L	<0.36	1.6	0.36	01/28/26 20:47	
PFHpS	ng/L	<0.42	1.6	0.42	01/28/26 20:47	
PFHxA	ng/L	<0.31	1.6	0.31	01/28/26 20:47	
PFHxS	ng/L	<0.43	1.6	0.43	01/28/26 20:47	
PFMBA	ng/L	<0.47	3.2	0.47	01/28/26 20:47	
PFMPA	ng/L	<0.60	3.2	0.60	01/28/26 20:47	
PFNA	ng/L	<0.32	1.6	0.32	01/28/26 20:47	
PFNS	ng/L	<0.44	1.6	0.44	01/28/26 20:47	
PFOA	ng/L	<0.46	1.6	0.46	01/28/26 20:47	
PFOS	ng/L	<0.53	1.6	0.53	01/28/26 20:47	
PFOSA	ng/L	<0.31	1.6	0.31	01/28/26 20:47	
PFPeA	ng/L	<0.44	3.2	0.44	01/28/26 20:47	
PFPeS	ng/L	<0.26	1.6	0.26	01/28/26 20:47	

Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.

REPORT OF LABORATORY ANALYSIS

This report shall not be reproduced, except in full,
without the written consent of Pace Analytical Services, LLC.

Date: 02/04/2026 03:20 PM



QUALITY CONTROL DATA

Project: Effluent PFAS

Pace Project No.: 50422857

METHOD BLANK: 5460850

Matrix: Water

Associated Lab Samples: 50422857001

Parameter	Units	Blank Result	Reporting Limit	MDL	Analyzed	Qualifiers
PFTeDA	ng/L	<0.27	1.6	0.27	01/28/26 20:47	
PFTrDA	ng/L	<0.32	1.6	0.32	01/28/26 20:47	
PFUnA	ng/L	<0.35	1.6	0.35	01/28/26 20:47	
13C2-PFDoA (S)	%	41	10-130		01/28/26 20:47	
13C2-PFTA (S)	%	39	10-130		01/28/26 20:47	
13C24:2FTS (S)	%	50	40-200		01/28/26 20:47	
13C26:2FTS (S)	%	43	40-200		01/28/26 20:47	
13C28:2FTS (S)	%	44	40-300		01/28/26 20:47	
13C3-PFBS (S)	%	42	40-135		01/28/26 20:47	
13C3-PFHxS (S)	%	41	40-130		01/28/26 20:47	
13C3-PFPrA (S)	%	41	5-130		01/28/26 20:47	
13C3HFPO-DA (S)	%	38	40-130		01/28/26 20:47	ES0
13C4-PFBA (S)	%	40	5-130		01/28/26 20:47	
13C4-PFHpA (S)	%	39	40-130		01/28/26 20:47	ES0
13C5-PFHxA (S)	%	38	40-130		01/28/26 20:47	ES0
13C5-PFPeA (S)	%	39	40-130		01/28/26 20:47	ES0
13C6-PFDA (S)	%	39	40-130		01/28/26 20:47	ES0
13C7-PFUdA (S)	%	42	30-130		01/28/26 20:47	
13C8-PFOA (S)	%	39	40-130		01/28/26 20:47	ES0
13C8-PFOS (S)	%	38	40-130		01/28/26 20:47	ES0
13C8-PFOSA (S)	%	29	40-130		01/28/26 20:47	ES0
13C9-PFNA (S)	%	39	40-130		01/28/26 20:47	ES0
d3-MeFOSAA (S)	%	35	40-170		01/28/26 20:47	ES0
d3-NMeFOSA (S)	%	23	10-130		01/28/26 20:47	
d5-EtFOSAA (S)	%	34	25-135		01/28/26 20:47	
d5-NEtFOSA (S)	%	24	10-130		01/28/26 20:47	
d7-NMeFOSE (S)	%	26	10-130		01/28/26 20:47	
d9-NEtFOSE (S)	%	26	10-130		01/28/26 20:47	

LABORATORY CONTROL SAMPLE: 5460851

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
11Cl-PF3OUdS	ng/L	150	119	79	55-160	
3:3 FTCA	ng/L	198	136	69	65-130	
4:2 FTS	ng/L	150	127	84	70-145	
5:3 FTCA	ng/L	992	732	74	70-135	
6:2 FTS	ng/L	154	127	83	65-155	
7:3 FTCA	ng/L	992	672	68	50-145	
8:2 FTS	ng/L	154	126	82	60-150	
9Cl-PF3ONS	ng/L	150	137	91	70-155	
ADONA	ng/L	150	131	87	65-145	
HFPO-DA	ng/L	160	129	80	70-140	

Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.

REPORT OF LABORATORY ANALYSIS

This report shall not be reproduced, except in full,
without the written consent of Pace Analytical Services, LLC.

Date: 02/04/2026 03:20 PM



QUALITY CONTROL DATA

Project: Effluent PFAS

Pace Project No.: 50422857

LABORATORY CONTROL SAMPLE: 5460851

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
NEtFOSA	ng/L	38.4	32.2	84	65-145	
NEtFOSAA	ng/L	38.4	33.6	88	70-145	
NEtFOSE	ng/L	384	328	85	70-135	
NFDHA	ng/L	80	66.6	83	50-150	
NMeFOSA	ng/L	38.4	30.7	80	60-150	
NMeFOSAA	ng/L	38.4	31.8	83	50-140	
NMeFOSE	ng/L	384	330	86	70-145	
PFBA	ng/L	160	128	80	70-140	
PFBS	ng/L	35.2	28.7	81	60-145	
PFDA	ng/L	38.4	33.7	88	70-140	
PFDaA	ng/L	38.4	33.4	87	70-140	
PFDoS	ng/L	38.4	24.7	64	50-145	
PFDS	ng/L	38.4	31.7	82	60-145	
PFEESA	ng/L	70.4	57.1	81	70-140	
PFHpA	ng/L	38.4	32.5	85	70-150	
PFHpS	ng/L	38.4	31.2	81	70-150	
PFHxA	ng/L	38.4	32.1	84	70-145	
PFHxS	ng/L	35.2	26.7	76	65-145	
PFMBA	ng/L	80	65.7	82	60-150	
PFMPA	ng/L	80	64.4	80	55-140	
PFNA	ng/L	38.4	31.5	82	70-150	
PFNS	ng/L	38.4	32.2	84	65-145	
PFOA	ng/L	38.4	31.8	83	70-150	
PFOS	ng/L	38.4	29.9	78	55-150	
PFOSA	ng/L	38.4	33.2	86	70-145	
PFPeA	ng/L	80	64.3	80	65-135	
PFPeS	ng/L	38.4	28.8	75	65-140	
PFTeDA	ng/L	38.4	34.1	89	60-140	
PFTrDA	ng/L	38.4	32.3	84	65-140	
PFUnA	ng/L	38.4	31.7	83	70-145	
13C2-PFDaA (S)	%			86	10-130	
13C2-PFTA (S)	%			77	10-130	
13C24:2FTS (S)	%			114	40-200	
13C26:2FTS (S)	%			99	40-200	
13C28:2FTS (S)	%			96	40-300	
13C3-PFBS (S)	%			102	40-135	
13C3-PFHxS (S)	%			102	40-130	
13C3-PFPrA (S)	%			107	5-130	
13C3HFPO-DA (S)	%			101	40-130	
13C4-PFBA (S)	%			105	5-130	
13C4-PFHpA (S)	%			100	40-130	
13C5-PFHxA (S)	%			103	40-130	
13C5-PFPeA (S)	%			103	40-130	

Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.

REPORT OF LABORATORY ANALYSIS

This report shall not be reproduced, except in full,
without the written consent of Pace Analytical Services, LLC.

Date: 02/04/2026 03:20 PM



QUALITY CONTROL DATA

Project: Effluent PFAS
Pace Project No.: 50422857

LABORATORY CONTROL SAMPLE: 5460851

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
13C6-PFDA (S)	%.			102	40-130	
13C7-PFUdA (S)	%.			102	30-130	
13C8-PFOA (S)	%.			101	40-130	
13C8-PFOS (S)	%.			95	40-130	
13C8-PFOSA (S)	%.			78	40-130	
13C9-PFNA (S)	%.			103	40-130	
d3-MeFOSAA (S)	%.			91	40-170	
d3-NMeFOSA (S)	%.			65	10-130	
d5-EtFOSAA (S)	%.			86	25-135	
d5-NEtFOSA (S)	%.			66	10-130	
d7-NMeFOSE (S)	%.			70	10-130	
d9-NEtFOSE (S)	%.			70	10-130	

Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.

REPORT OF LABORATORY ANALYSIS

This report shall not be reproduced, except in full,
without the written consent of Pace Analytical Services, LLC.



QUALIFIERS

Project: Effluent PFAS
Pace Project No.: 50422857

DEFINITIONS

DF - Dilution Factor, if reported, represents the factor applied to the reported data due to dilution of the sample aliquot.

ND - Not Detected at or above adjusted reporting limit.

TNTC - Too Numerous To Count

J - Estimated concentration above the adjusted method detection limit and below the adjusted reporting limit.

MDL - Adjusted Method Detection Limit.

PQL - Practical Quantitation Limit.

RL - Reporting Limit - The lowest concentration value that meets project requirements for quantitative data with known precision and bias for a specific analyte in a specific matrix.

S - Surrogate

1,2-Diphenylhydrazine decomposes to and cannot be separated from Azobenzene using Method 8270. The result for each analyte is a combined concentration.

Consistent with EPA guidelines, unrounded data are displayed and have been used to calculate % recovery and RPD values.

LCS(D) - Laboratory Control Sample (Duplicate)

MS(D) - Matrix Spike (Duplicate)

DUP - Sample Duplicate

RPD - Relative Percent Difference

NC - Not Calculable.

SG - Silica Gel - Clean-Up

U - Indicates the compound was analyzed for, but not detected.

N-Nitrosodiphenylamine decomposes and cannot be separated from Diphenylamine using Method 8270. The result reported for each analyte is a combined concentration.

Reported results are not rounded until the final step prior to reporting. Therefore, calculated parameters that are typically reported as "Total" may vary slightly from the sum of the reported component parameters.

Pace Analytical is TNI accredited. Contact your Pace PM for the current list of accredited analytes.

TNI - The NELAC Institute.

ANALYTE QUALIFIERS

ES0 Extracted Internal Standard recovery outside laboratory control limits.

I The ion ratio is outside control limits.

REPORT OF LABORATORY ANALYSIS

This report shall not be reproduced, except in full,
without the written consent of Pace Analytical Services, LLC.

Date: 02/04/2026 03:20 PM



QUALITY CONTROL DATA CROSS REFERENCE TABLE

Project: Effluent PFAS
Pace Project No.: 50422857

Lab ID	Sample ID	QC Batch Method	QC Batch	Analytical Method	Analytical Batch
50422857001	Effluent	EPA 1633	1049892	EPA 1633	1050261



REPORT OF LABORATORY ANALYSIS

This report shall not be reproduced, except in full,
without the written consent of Pace Analytical Services, LLC.



Sample Conditions Upon Receipt Form (SCUR)

Date/Time: 1.22.26	Evaluated By: <i>WTE</i>	WO#: 50422857	
Client: City of Cadillac	PM: BJH	Due Date: 02/13/26 CLIENT: GR-Cadillac	
Lab Notified of Rush or Short Holds:	YES NO NA		
Project Received Via: FedEx UPS Client Pace Courier Other: _____	Comments:		
Custody Seal Present and Intact:	YES NO NA		
Received Sample Information Form (SIF): Drinking Waters Only	YES NO NA		
Short Hold Present (≤ 48 Hours):	YES NO NA		
Sample Received in Hold:	YES NO NA		
Custody Signature Present:	YES NO NA		
Collector Signature Present:	YES NO NA		
Sample Collected Today and On Ice:	YES NO N/A		
IR Gun #: 350 351 Therm #: 282 283	Temp. should be 0°C - 6°C (Initial/Corrected)		
Ice Type: WET Bagged / WET Loose BLUE NONE	1. Cooler Temp. Upon Receipt: _____ °C		
Ice Location: TOP BOTTOM MIDDLE DISPERSED	2. Cooler Temp. Upon Receipt: _____ °C		
Temp Blank Received:	YES NO NA		
Sample Label Matches COC (ID/Date/Time):	YES NO NA		
Container Intact:	YES NO NA		
Correct Container:	YES NO NA		
Sufficient Volume:	YES NO NA		
Sample pH Acceptable: All containers needing preservation are found to be in compliance with EPA recommendation. Denote with pink sticker on the lid if unacceptable. pH Strip Lot #: _____ Exceptions are VOA, coliform, LLHg, O&G/TPH, or any container with a septum cap or preserved with HCl	YES NO N/A		
Residual Chlorine Absent: Cl ₂ Strip Lot #: _____ Applies to SVOC 625, PCB/Pest. 608, Total/Amenable/Available/Free Cyanide	YES NO N/A		
VOA Headspace Acceptable (<6mm): Denote with silver x on rim of container cap if unacceptable.	YES NO N/A		
Trip Blank Received: HCl MeOH Other: _____	YES NO ON HOLD		
Comments:	3. Cooler Temp. Upon Receipt: _____ °C		
	4. Cooler Temp. Upon Receipt: _____ °C		
	Non-Conformance Form Required: YES NO		